HireX

AIPowered Recruitment and Hiring Platform

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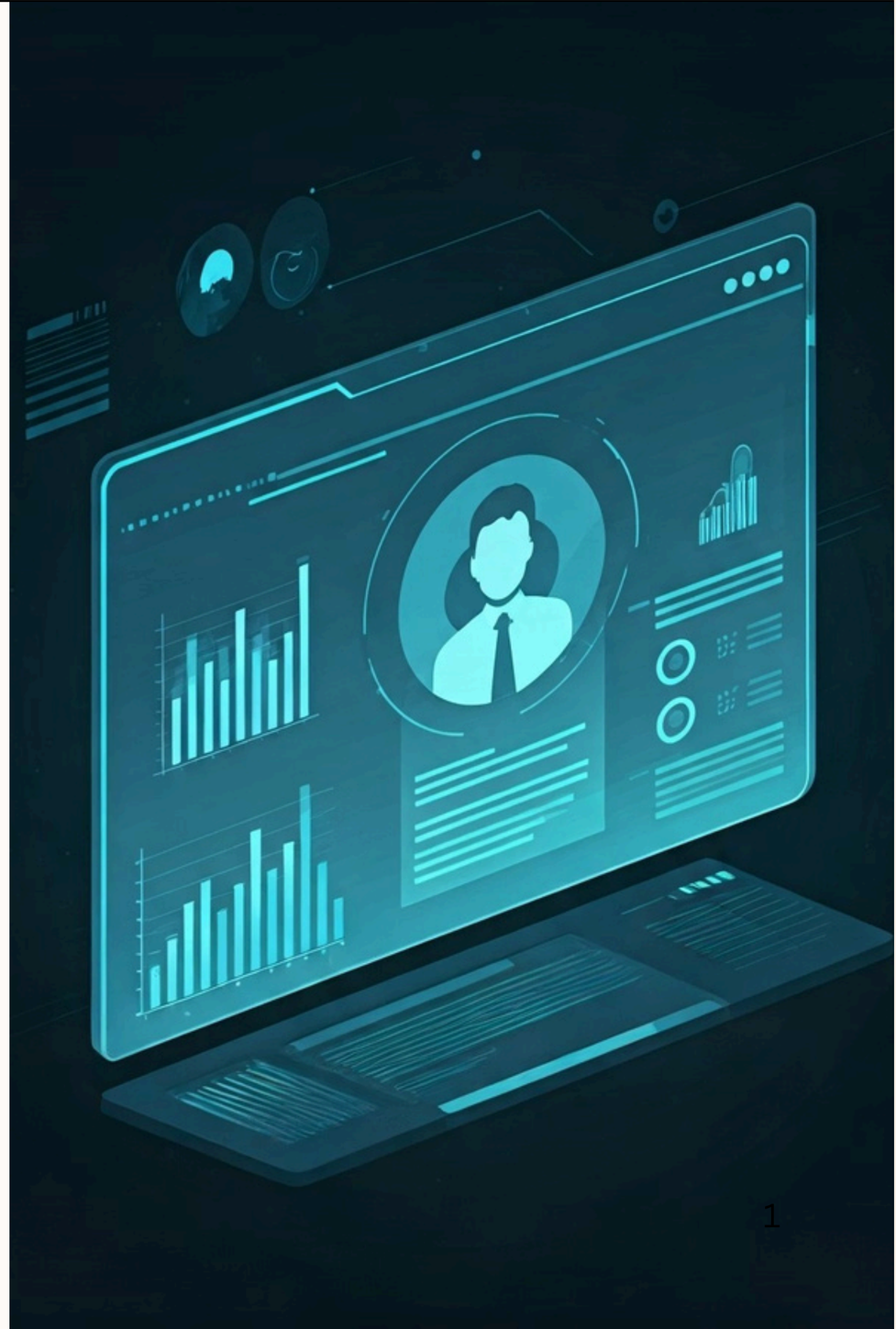


Motivation

- Recruitment platforms receive a high volume of applications for each job posting, making manual screening inefficient.
- Recruiters spend significant time reviewing resumes, leading to delayed hiring decisions.
- Traditional recruitment systems rely heavily on keyword-based filtering, resulting in poor candidate–job matching.
- Skill mismatch between candidates and job requirements reduces hiring quality.
- Manual evaluation processes are prone to bias and inconsistency.
- There is a growing need for intelligent systems that automate screening, ensure fairness, and improve recruitment accuracy.

Objectives

01

Automated Recruitment Platform

Design and develop HireX as an AI-powered system that automates key recruitment activities from job posting to candidate shortlisting.

02

Intelligent Candidate–Job Matching

Improve matching accuracy by analyzing resumes and job requirements using skill extraction and AI-based matching techniques.

03

Efficiency and Time Optimization

Reduce recruiter workload and hiring time by automating resume screening, ranking, and interview-related processes.

04

Strategic Recruitment Management System

Enable efficient recruitment through intelligent automation, structured workflows, and data-driven decision support to improve hiring outcomes.

Scope of the Project

1

User and Role Management

Support multiple user roles, including job seekers, recruiters, and administrators, with secure authentication and role-based access.

2

AI-Based Candidate Evaluation

Apply resume parsing, skill extraction, matching, ranking, and shortlisting to improve candidate evaluation accuracy.

3

Recruitment Workflow Automation

Manage the complete application lifecycle, from job search and application submission to screening and interview scheduling.

4

Scalable and Extensible Architecture

Design the system to support ATS automation, analytics, and advanced AI enhancements without major architectural changes.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Person-job fit estimation... with Co-Attention Neural Networks [1]	1. **mashRNN Encoding**: Hierarchical RNNs extract semantic features from long-form documents at multiple levels. 2. **Co-Attention Mechanism**: Aligns specific job requirements with resume experiences to quantify matching contribution. 3. **Graph Neural Networks (GNN)**: Models "global recruiter experience" by aggregating historical successful recruitment records. 4. **BiLSTM with Attention**: Computes similarity to construct edges in job-resume graphs.	• **Captures Recruitment History**: Utilizes past successful hires to guide current predictions, modeling recruiter intuition. • **Handles Long Documents**: mashRNN effectively processes lengthy resumes unlike standard RNNs. • **High Interpretability**: Co-attention highlights exactly which skills matched which requirements.	• **High Computational Cost**: GNNs and hierarchical RNNs require significant processing power and training time. • **Data Dependency**: Requires a large dataset of historical successful matches to build effective graphs. • **Inference Latency**: Slower real-time performance compared to simpler models due to complex graph operations.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
AI-Powered Resume Screening System using NLP and Machine Learning [2]	1. **Grid-Search Tuned SVM**: Linear Support Vector Machine with hyperparameter optimization (C/Kernel) via cross-validation. 2. **TF-IDF Vectorization**: Converts text into a sparse matrix of 5000 features based on term importance weights. 3. **NLP Preprocessing**: Standardizes text via tokenization, stopword removal, and lemmatization.	• **High Classification Accuracy**: Achieved 91.3% accuracy through rigorous hyperparameter tuning. • **Computational Efficiency**: TF-IDF and Linear SVM are less resource-intensive than deep learning models. • **Strong Generalization**: Cross-validation ensures the model performs well on unseen data.	• **Lacks Semantic Context**: TF-IDF ignores word order and context, missing nuances in similar skills. • **Sparse Feature Matrix**: High-dimensional sparse data can lead to memory inefficiencies. • **Fixed Categories**: Limited to predicting predefined job roles; struggles with interdisciplinary profiles.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Intelligent Resume Evaluation Tool Based on Machine Learning [3]	1. **Weighted Resume Scoring**: Assigns quantifiable scores based on defined weights (e.g., Skills: 30, Education: 20). 2. **Skill Gap Analysis (PhraseMatcher)**: Uses spaCy to compare extracted skills against predefined industry lists. 3. **Regex & Entity Extraction**: patterns to clean text and extract structured entities (email, phone).	• **Actionable Feedback**: Provides specific scores and course recommendations to improve candidate profiles. • **Quantifiable Quality**: Converts qualitative resume content into a comparable numerical score. • **High Precision Extraction**: Regex and PhraseMatcher are highly accurate for structured data.	• **Static Dependencies**: Relies on predefined skill lists which may become outdated quickly. • **Subjective Weighting**: Manual assignment of scoring weights may not reflect actual market value. • **Format Sensitivity**: Parsing failures occur if resume formatting deviates from standard layouts.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Resume Ranker: AI-Based Skill Analysis and Skill Matching System [4]	1. **Contextual Keyword Extraction (KeyBERT)**: Uses BERT embeddings to extract context-aware keywords rather than simple frequency. 2. **Synonym Generation (Word2Vec)**: Identifies semantic relationships to capture synonyms and related terms. 3. **Cosine Similarity Matching**: Measures geometric alignment between candidate vectors and job descriptions.	• **Semantic Understanding**: Captures context and nuances, outperforming simple keyword matching. • **Handles Synonyms**: Word2Vec ensures candidates aren't penalized for using different terminology for the same skill. • **Visual Ranking**: Provides clear percentage-based matching scores for easy comparison.	• **High Computational Overhead**: BERT-based models require significant resources for inference. • **Complex Implementation**: Requires integrating and maintaining multiple NLP models (BERT, Word2Vec). • **Corpus Dependent**: Quality of synonyms depends heavily on the size and domain of the training corpus.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Streamlining Talent Acquisition: A Machine Learning Approach [5]	1. **Encrypted Data Processing**: Secures sensitive applicant data using **AES-256 encryption** protocols. 2. **Bias-Reduced Matching**: Uses data-driven skill criteria rather than demographic identifiers for ranking. 3. **NLP Parsing**: Automates the extraction of structured fields from unstructured resume text.	• **High Data Security**: Ensures compliance with privacy laws (GDPR) and protects applicant identity. • **Fair Assessment**: Reduces unconscious human bias by focusing purely on data points. • **Scalability**: Can handle fluctuating volumes of applications without compromising processing speed.	• **Processing Overhead**: Encryption/decryption steps add latency to the data pipeline. • **Key Management Complexity**: Requires robust infrastructure to manage encryption keys securely. • **Parsing Limitations**: Accuracy depends heavily on the structure of input documents (PDF/Docx).

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Recruiter's perception of artificial intelligence (AI)-based tools in recruitment [6]	1. **Extended UTAUT Framework:** Adapted the Unified Theory of Acceptance and Use of Technology by adding frequency of use and education. 2. **Mixed Methods Approach:** Combined quantitative survey data with qualitative coding of open-ended responses. 3. **Web-based Survey (Prolific Academic):** Data collection via a specialized platform for scientific research. 4. **Hierarchical Regression Analysis:** Statistical analysis using three blocks of variables to test predictors.	• **Holistic Prediction:** The extended UTAUT model effectively identifies performance expectancy as the strongest predictor of behavioral intention. • **Data Quality:** Using Prolific Academic ensures high transparency and internal reliability of psychometric scales. • **Nuanced Insight:** The mixed-method approach explains *why* recruiters value efficiency while fearing the lack of human judgment. • **Variable Isolation:** Hierarchical regression isolates the unique effect of variables.	• **Limited Generalizability:** The study relies on specific variables, potentially excluding factors like trust, security, or legal issues (GDPR). • **Self-Report Bias:** Survey data reflects perception rather than objective system performance or actual usage logs. • **Weak Demographics Link:** The analysis found that age, gender, and experience had no significant impact, suggesting these standard moderators may not apply in this specific context.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Evolution of Applicant Tracking Systems: From Databases to Intelligent Platforms [7]	1. **Comparative Analytical Method:** Synthesized model architectures and platform patterns from peer-reviewed literature. 2. **Embedding-Based Ranking (BERT/S-BERT):** Uses dense vector similarity and transformer encoders for semantic matching. 3. **Human-in-the-Loop Re-ranking:** Integrates recruiter feedback signals into the ranking pipeline.. 4. **Governance/Risk-Control Matrix:** Framework for bias audits, drift detection, and explainability.	• **Semantic Precision:** Transformer models (BERT) capture skill equivalence and synonyms better than keyword filters. • **Reduced Workload:** Intelligent triage compresses screening time, directly addressing high cost-per-hire and time-to-fill bottlenecks. • **Bias Mitigation:** Human-in-the-Loop and governance matrices make trade-offs inspectable and prevent "black box" decision-making. • **Scalability:** Event-driven architectures allow systems to handle applicant spikes without latency degradation.	• **Data Dependency:** Effectiveness relies heavily on the quality of talent corpora used to tune the encoders. • **Complexity:** Implementing continuous bias monitoring and modular inference graphs increases technical overhead compared to static filters. • **Historical Bias Risk:** If not carefully managed, the system may imprint historical inequities found in training data onto new rankings.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Smart Hiring Redefined: An Intelligent Recruitment Management Platform [8]	1. **J2EE Three-Tier Architecture:** Structured into presentation, business logic, and data service layers. 2. **MVC (Model-View-Controller) Pattern:** Separates business logic from user interface and data access. 3. **Role-Based Access Control (RBAC):** Manages permissions for students, enterprises, and administrators. 4. **Black Box Testing:** Validates functionality against requirements without inspecting internal code paths.	• **Maintainability:** The layered architecture ensures high cohesion and low coupling, making future modifications easier. • **User-Centric Validation:** Black box testing simulates real-world user behavior to ensure workflows (like resume submission) function as expected. • **Data Integrity:** Database normalization (foreign keys) ensures consistent relationships between colleges, majors, and students.	• **Architectural Coupling:** In some implementations, the service layer directly manipulated request objects, reducing true separation of concerns. • **Performance Limits:** The current design may struggle with high concurrency processing and lacks asynchronous processing for heavy loads. • **Mobile Limitations:** The system currently lacks mobile device adaptation, which restricts accessibility for students.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Enhancing recruitment efficiency: An advanced Applicant Tracking System (ATS) [9]	1. **Natural Language Processing (NLP):** Uses TF-IDF and tokenization for text parsing and cleaning. 2. **K-Nearest Neighbors (KNN):** Algorithm used for classifying resumes based on skill proficiency and experience. 3. **Agile Development Methodology:** Utilizes Scrum/Kanban for iterative development and deployment. 4. **Cloud-Based Deployment:** Architecture designed for scalability via cloud computing resources.	• **Structured Data Extraction:** NLP effectively converts unstructured resume text into structured formats for analysis. • **Pattern Recognition:** The system handles complex relationships and patterns in data without assuming underlying distributions. • **Rapid Adaptation:** Agile methodology enables the team to quickly incorporate feedback and release features incrementally.	• **Computational Cost:** The KNN algorithm became computationally inefficient when processing large datasets. • **Infrastructure Overhead:** While scalable, cloud infrastructure entails management overhead and potential cost implications. • **Integration Complexity:** Merging the ATS with existing infrastructures presents obstacles regarding data transfer and compatibility.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
A Multi-Module Applicant Tracking System: From Job Posting To Employee Performance Monitoring [10]	1. **Modular Architecture**: System divided into independent Hiring, Employee Management, and Performance modules. 2. **MERN Stack Implementation**: Uses MongoDB, Express, React, and Node.js for development. 3. **Rule-Based & ML Scoring**: Combines machine learning with rule-based logic for candidate ranking. 4. **Agile/Iterative Development**: employed sprint cycles with continuous user feedback.	• **End-to-End Management**: Covers the entire lifecycle from job posting to performance analytics, bridging the gap between ATS and HRMS. • **Extensibility**: The modular design allows specific components (like performance tracking) to be updated without affecting others. • **Efficiency**: Users reported a greater than 60% reduction in manual effort due to automation features.	• **Generalizability Limits**: Early iterations or specific algorithm choices may lack generalizability across different industries. • **External Factors**: Data-centric performance modules may overlook external, qualitative factors influencing employee performance. • **Dependency on Feedback:** The development relies heavily on test user feedback, which might bias the system toward specific user preferences.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Enhancing Talent Search Ranking with Role-Aware Expert Mixtures and LLM-based Fine-Grained Job Descriptions [11]	1. **Fine-Grained JD Encoding:** Uses LLMs with Chain-of-Thought (CoT) prompting to extract implicit preferences,. 2. **Gated Cross Network (GCN):** Models interactions between JD and resume embeddings. 3. **Role-Aware MMoE:** Multi-gate Mixture-of-Experts network with role-specific gating 4. **Multi-Task Learning (MTL):** Jointly predicts CTR, CVR, and resume relevance using a shared expert architecture. 5. **Online A/B Testing:** Evaluated using Google's standard protocol.	• **JD Encoding:** Captures implicit signals and fine-grained preferences often missed by keyword matching. • **Role-Aware MMoE:** Adapts ranking strategies to the behavioral heterogeneity of different recruiter roles. • **MTL:** Mitigates behavioral noise from subjective judgments and improves robustness of predictions,. • **Impact:** Demonstrated significant gains in CVR (5.97%) and Click-Through Conversion Rate (17.29%),.	• **Data Dependency:** Relies on sufficient historical interaction data, limiting effectiveness in cold-start scenarios. • **Temporal Limitations:** Does not explicitly account for temporal dynamics in recruiter behavior or job market trends. • **Complexity:** Increasing the number of experts beyond three resulted in decreased performance due to overfitting.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Fairness-Aware Ranking in Search & Recommendation Systems with Application to LinkedIn Talent Search [12]	1. **Bias Quantification Measures:** Skew@k, MinSkew, MaxSkew, and NDKL 2. **DetGreedy Algorithm:** Score maximizing greedy mitigation for re-ranking,. 3. **DetCons Algorithm:** Greedy conservative mitigation focusing on minimum representation requirements,. 4. **DetRelaxed Algorithm:** A relaxed variant of DetCons using integrality constraints. 5. **DetConstSort Algorithm:** Interval constrained sorting for feasible mitigation.	• **DetGreedy:** Simple to implement, high utility maintenance, and efficient for attributes with low cardinality. • **DetConstSort:** Theoretically proven to be feasible for satisfying fairness constraints,. • **Framework:** Achieves fairness criteria like equality of opportunity and demographic parity without statistically significant loss in business metrics,,.	• **DetGreedy:** Not guaranteed to be feasible when the number of attribute values is $\ge$ 4. • **NDKL:** Cannot differentiate between bias of equal extent in opposite directions and is hard to interpret. • **Skew@k:** Defined for single attributes and depends on the choice of k.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
AI-Driven Decision-Making System for Hiring Process [13]	1. **Modular Multi-Agent Pipeline:** Sequential agents for ingestion, context construction, verification, and scoring. 2. **Public Data Verification:** Agents use MCP tools to scrape and verify data from platforms like LinkedIn and GitHub. 3. **Entity Linking (NER+ED+NEL):** Extracts and normalizes technologies/skills from heterogeneous inputs. 4. **Scoring Formula:** Combines technical fit, culture fit, and normalized risk penalties..	• **Efficiency:** Reduces screening time per qualified candidate from 3.33 hours (expert human) to 1.70 hours. • • **Explainability:** Produces traceable component-level reports rather than just a black-box score. • **Cost:** Significantly lower estimated processing cost ($2.29) compared to human recruiters (~$50-$95). • **Risk Management:** Explicitly penalizes risks (e.g., date inconsistencies, toxic content) in the final score.	• **Bias Inheritance:** Inherits biases present in the recruiters' judgments used as ground truth for evaluation. • • **Ethical/Privacy Concerns:** Public data scraping raises issues regarding privacy, consent, and potential bias. • • **Cost Dependency:** Operational costs are tied to third-party API pricing (OpenAI).

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Optimizing Job Matching Through Automated Resume Screening with NLP and Machine Learning [14]	1. **Preprocessing Pipeline:** Remove URLs, hashtags, mentions, and punctuation. 2. **Text Normalization:** Tokenization, Stemming, and Lemmatization. 3. **TF-IDF Vectorization:** Converts cleaned text into numerical vectors based on term importance. 4. **Machine Learning Classification:** Compare KNN, Random Forest, SVM, and Logistic Regression. 5. **One-vs-Rest Classification:** Adapted Logistic Regression for multi-class categorization.	• **SVM Performance:** Support Vector Machine achieved the highest accuracy (1.0 training, ~0.99 testing). • **TF-IDF:** Effectively highlights important terms relative to the corpus, filtering out common but irrelevant words. • **Efficiency:** Automates the burdensome manual sorting process, allowing HR to focus on interviews.	• **KNN Performance:** KNN demonstrated the lowest accuracy among the tested models. • **Stemming Limitations:** Stemming is less accurate than lemmatization as it simply chops word ends rather than using a dictionary. • **Complexity:** Requires extensive preprocessing (cleaning, tokenizing) before the model can process data.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
ResumAI: Revolutionizing Automated Resume Analysis and Recommendation with Multi-Model Intelligence [15]	1. **Named Entity Recognition (NER):** Uses SpaCy to extract names and specific entities,. 2. **Resume Parsing:** Converts files with pdfminer /doc2text and extracts contacts via regex/URLs. 3. **Similarity Computation:** Ranks using cosine similarity on BERT and TF-IDF embeddings. 4. **Classification Models:** Comparison of Multinomial Naive Bayes, Linear SVC, and k-NN classifiers,.	• **Linear SVC Accuracy:** The Linear Support Vector Classifier achieved a high accuracy of 98.63%. • • **Contextual Analysis:** BERT embeddings capture contextual relationships between words, providing rich representations. • **Automation:** Eliminates manual crawling by extracting structured data from unstructured formats (PDF/DOCX),.	• **Computation:** BERT is computationally complex compared to simpler methods like TF-IDF. • **Naive Bayes Performance:** Multinomial Naive Bayes showed significantly lower accuracy (73.97%) compared to SVC. • **Dependency:** Requires vectorization of textual data before any ML processing can occur.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Predicting Skill Shortages in Labour Markets: A Machine Learning Approach [16]	1. **Data Integration:** Linked job ads with labour statistics using ANZSCO taxonomy. 2. **RCA Metric:** Identified emerging skills via normalized importance. 3. **XGBoost Classification:** Predicted skill shortage status. 4. **Lagged Features:** Captured temporal trends using past data.	• **Granular Insight:** RCA reveals specialised/emerging skills that frequency counts miss • **High Performance:** XGBoost handles missing data well and achieved up to 83% Macro-F1 scores,. • **Scalability:** Uses standard labour demand/supply data available in most advanced economies, making it replicable. • **Interpretability:** Feature importance confirms economic theories	• **Inertia Bias:** Auto-regressive features dominate predictions, making the model poor at predicting *changes* or "flips" in shortage status,. • **Data Representativeness:** Online job ads over-represent white-collar roles and miss informal/traditional hiring sectors. • **Class Imbalance:** Shortage events are rare, requiring synthetic oversampling which can lead to overfitting,.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
A Comprehensive Survey of Artificial Intelligence Techniques for Talent Analytics [17]	1. **Taxonomy Creation:** Categorised AI use cases in HR and labour markets. 2. **Parsing:** NLP/OCR extract structured resume and job data. 3. **Person–Job Fit:** Deep models match candidates to roles. 4. **ONA:** Graph models analyse employee networks. 5. **Forecasting:** Time-series models predict skill demand trends.	• **Semantic Depth:** BERT/GNN capture rich contextual meaning beyond keywords. • **Objectivity:** ONA reduces bias versus manual evaluations. • **Efficiency:** Automates resume screening and benchmarking at scale. • **Dynamic Modelling:** Tracks evolving skills using time-series models.	• **Data Hunger:** Requires large, labeled HR datasets. • **Bias Risk:** Can amplify historical gender or racial biases. • **Low Interpretability:** Black-box models limit explainability. • **Privacy Concerns:** ONA raises ethical and employee privacy issues.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Tec-Habilidad: Skill Classification for Bridging Education and Employment [18]	1. **Zero-shot Extraction:** Used GPT-3.5-Turbo to extract skills without training data. 2. **KSAO Mapping:** Grouped skills into Knowledge, Skills, Abilities, and Other. 3. **Definition Augmentation:** Added skill definitions to improve context. 4. **Zero-shot Classification:** Evaluated LLMs for skill classification tasks.	• **Language Gap:** Addresses lack of non-English (Spanish) skill datasets. • **Definition Boost:** Adding definitions significantly improved accuracy. • **Automation Potential:** Fine-tuned Transformers outperformed human agreement.	• **LLM Limitations:** Zero-shot LLMs performed significantly worse (0.40-0.48 accuracy) compared to fine-tuned BERT models (0.85+),. • **Soft Skill Nuance:** LLMs struggled to identify implicit soft skills, achieving only 52% overlap with human labels. • **Class Confusion:** Models struggled to distinguish between "Skill" and "Knowledge" due to conceptual overlap,.

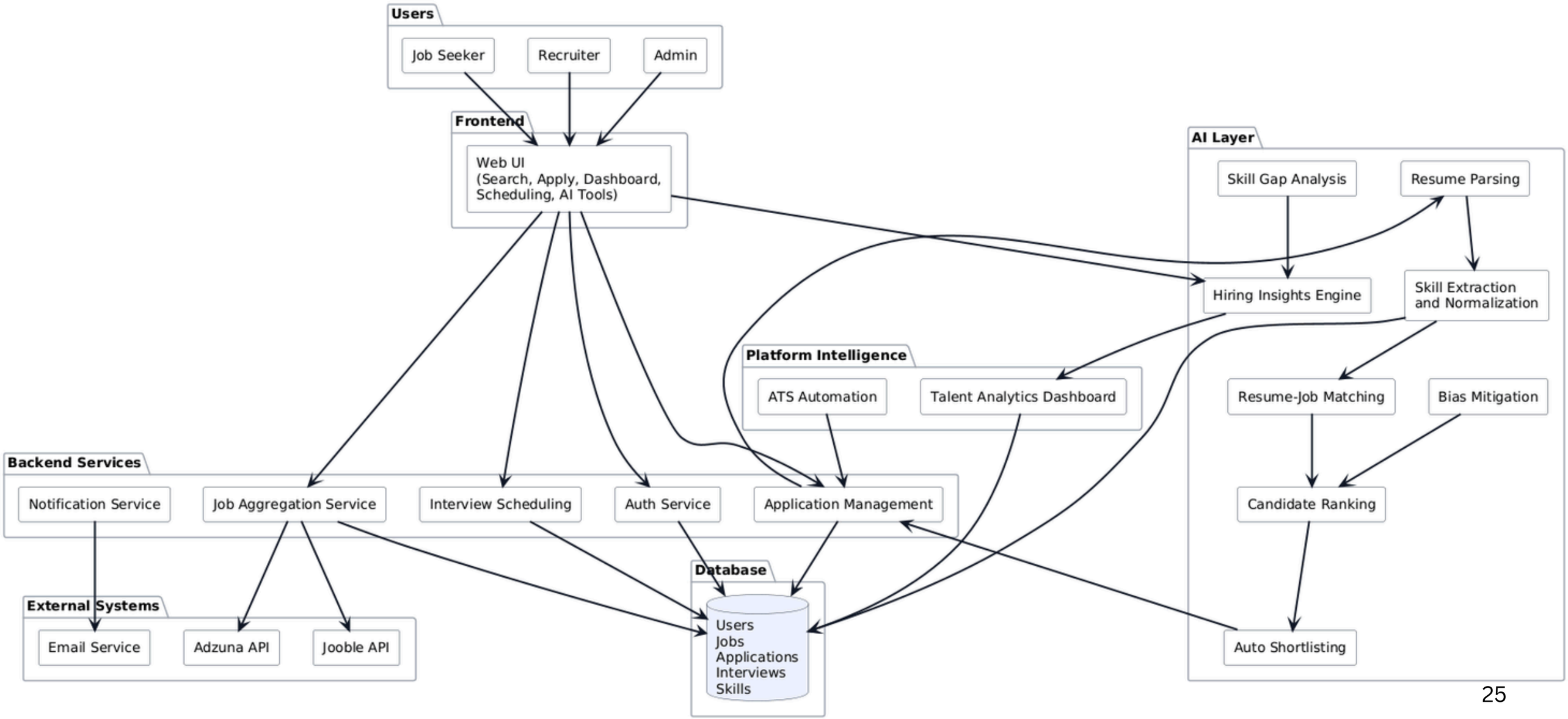
Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Real-Time AI-Based Skill Gap Analysis and Adaptive Career Guidance [19]	1. **Feature Encoding:** Generated NLP embeddings for candidates and jobs. 2. **Similarity Scoring:** Weighted cosine similarity on skills and experience. 3. **Generative Matching:** Used Gemini Flash 1.5 for semantic matching. 4. **Skill Gap Detection:** Classified missing skills as critical or recommended. 5. **Market Intelligence:** Integrated real-time labour trend data.	• **Integrity Assurance:** Prevents impersonation via facial recognition. • **Personalisation:** Recommends targeted projects and courses. • **Holistic Matching:** Weighs verified and inferred skills. • **Dynamic Updates:** Auto-updates profiles after tests.	• **Complexity:** Managing weights for similarity scoring (experience vs. certs vs. skills) introduces complexity in tuning the matching algorithm. • **Dependency:** Relies heavily on external APIs (Gemini, News API), creating potential points of failure or cost scaling issues. • **Input Quality:** Profile accuracy depends on comprehensive user input; poor data entry degrades matching quality.

Literature Survey

PAPER TITLE	METHODOLOGY	ADVANTAGES	DISADVANTAGES
Using Graph Algorithms for Skills Gap Analysis [20]	1. **Knowledge Graph:** Integrated employee data with O*NET in Neo4j. 2. **PageRank:** Identified important skills and tasks globally and per role. 3. **Node Similarity:** Linked employees using Jaccard similarity. 4. **Community Detection:** Used Louvain to find functional groups. 5. **Triangle Count:** Measured network cohesion and stability.	• **Standardisation:** Integrating O*NET data provides a transferable, standardised ontology for skills across the organisation. • **Hidden Connections:** Identifies employees with similar skills who aren't formally connected, aiding internal mobility/recruiting. • **Network Stability:** Triangle count metrics help identify fragile teams or isolated employees needing better integration. • **Targeted Prioritisation:** Personalised PageRank helps HR prioritise which skills are most critical when hiring for roles.	• **Data Sensitivity:** Graph connectivity is highly sensitive to the chosen similarity cutoff threshold (e.g., 0.5), which requires business domain tuning. • **Interpretation Difficulty:** Large communities detected by Louvain can be generic and hard to interpret without deep domain knowledge. • **Cold Start:** Requires a densely connected network; sparse data prevents effective community detection or similarity scoring.

Block Diagram



Implementation

Frontend

- React 18 with Vite
- React Router DOM
- TailwindCSS with Autoprefixer
- Lucide-React (UI icons)
- Recharts (data visualization)
- Axios (API communication)

Backend

- Node.js with Express.js
- JavaScript (ES6 Modules)
- RESTful API architecture
- JWT authentication and bcrypt.js
- Multer for resume file uploads

AI and Data Processing

- Natural.js for NLP and TF-IDF
- PDF resume parsing using pdf-parse
- Cosine similarity for matching
- Rule-based candidate ranking
- Break-aware round-robin scheduling algorithm

Implementation

Database

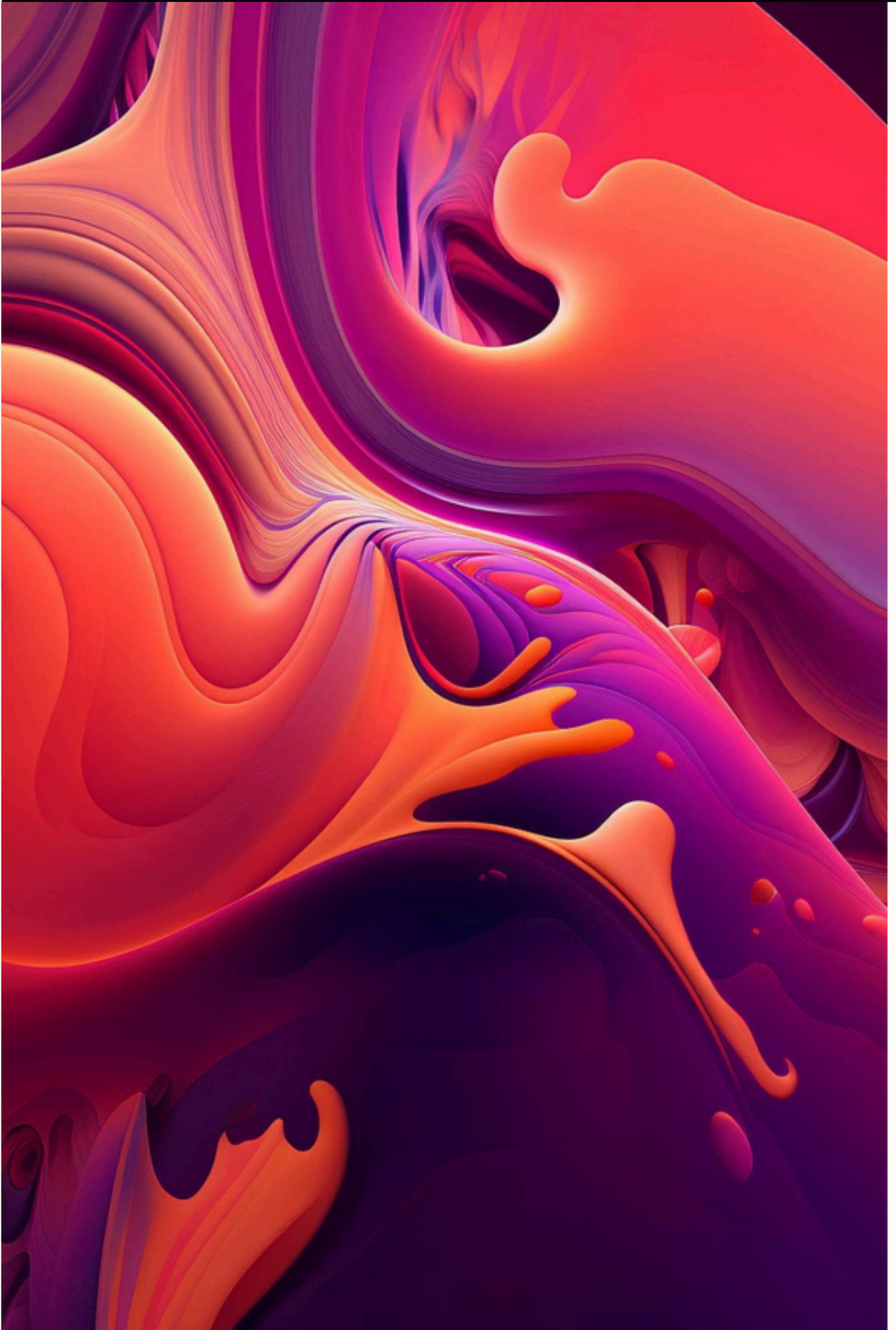
- PostgreSQL relational database
- Normalized schema design
- Core entities: Users, Jobs, Applications, Resumes, Interviews, Companies

External Integrations

- Adzuna API and Jooble API
- Notification service

Development Tools

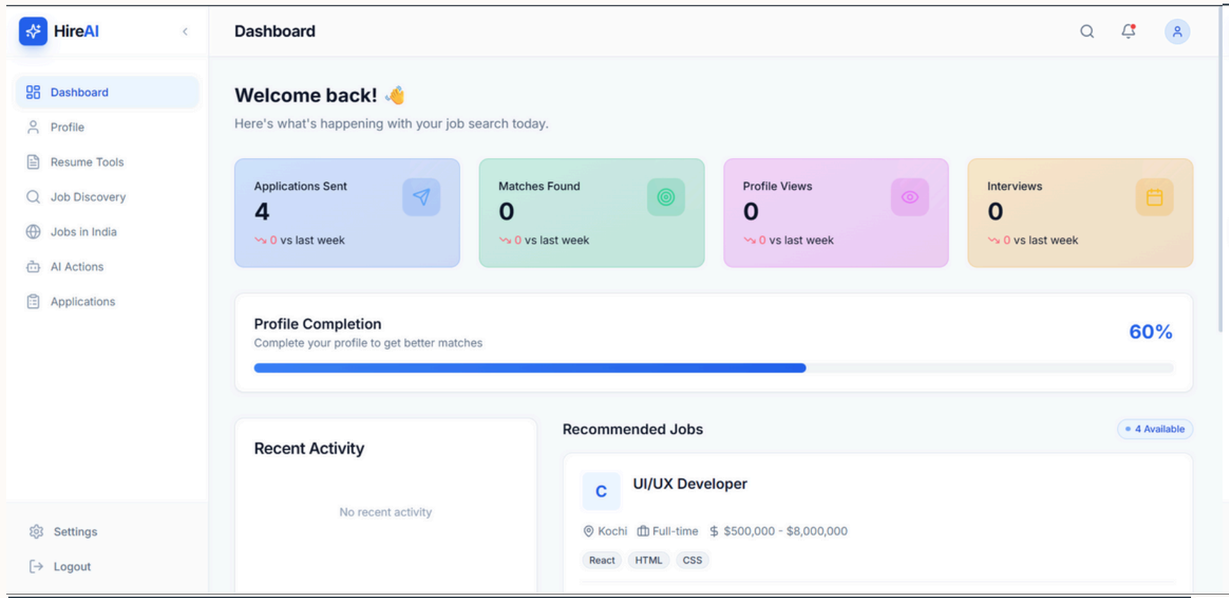
- Git for version control
- npm for package management
- Vite and Nodemon for development
- dotenv for environment management

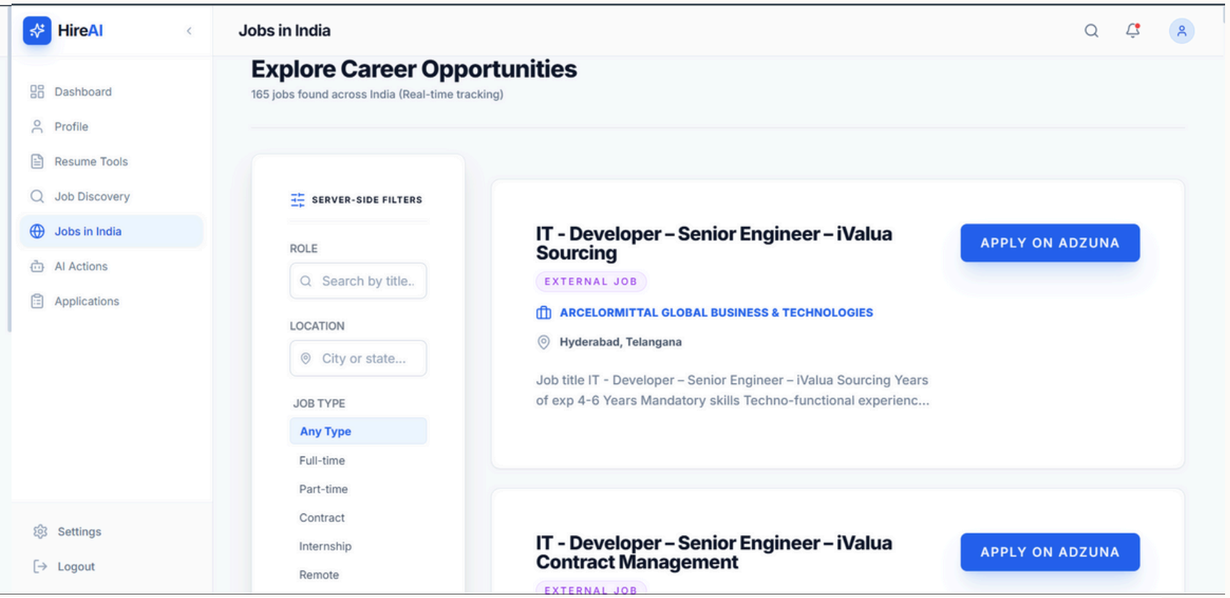


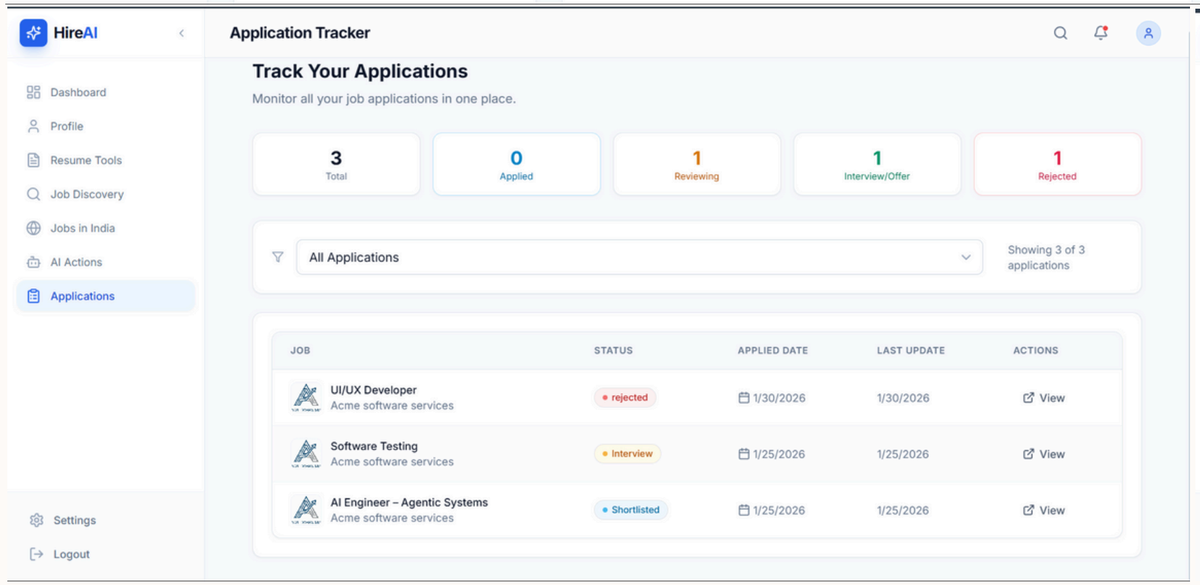
System Architecture

- HireX follows a layered, modular architecture to ensure scalability and maintainability.
- Users access the system through a unified frontend interface.
- The frontend communicates with backend services using secure APIs.
- Backend services manage authentication, job aggregation, applications, and scheduling.
- An AI layer performs resume parsing, matching, ranking, and shortlisting.
- A centralized database stores all recruitment-related data.
- External systems provide job listings and notification support.

Result


HireAI
Dashboard
Dashboard
Profile
Resume Tools
Job Discovery
Jobs in India
AI Actions
Applications
Settings
Logout
Welcome back! 👋
Here's what's happening with your job search today.
Applications Sent
4
0 vs last week
Matches Found
0
0 vs last week
Profile Views
0
0 vs last week
Interviews
0
0 vs last week
Profile Completion
Complete your profile to get better matches
60%
Recent Activity
No recent activity
Recommended Jobs
4 Available
C
UI/UX Developer
Kochi
Full-time
$500,000 - $8,000,000
React
HTML
CSS


HireAI
Jobs in India
Dashboard
Profile
Resume Tools
Job Discovery
Jobs in India
AI Actions
Applications
Settings
Logout
Explore Career Opportunities
165 jobs found across India (Real-time tracking)
SERVER-SIDE FILTERS
ROLE
Search by title..
LOCATION
City or state...
JOB TYPE
Any Type
Full-time
Part-time
Contract
Internship
Remote
IT - Developer – Senior Engineer – iValua Sourcing
EXTERNAL JOB
ARCELORMITTAL GLOBAL BUSINESS & TECHNOLOGIES
Hyderabad, Telangana
Job title IT - Developer – Senior Engineer – iValua Sourcing Years of exp 4-6 Years Mandatory skills Techno-functional experienc...
APPLY ON ADZUNA
IT - Developer – Senior Engineer – iValua Contract Management
APPLY ON ADZUNA


HireAI
Application Tracker
Dashboard
Profile
Resume Tools
Job Discovery
Jobs in India
AI Actions
Applications
Settings
Logout
Track Your Applications
Monitor all your job applications in one place.
3
Total
0
Applied
1
Reviewing
1
Interview/Offer
1
Rejected
All Applications
Showing 3 of 3 applications
JOB
STATUS
APPLIED DATE
LAST UPDATE
ACTIONS
UI/UX Developer
Acme software services
rejected
1/30/2026
1/30/2026
View
Software Testing
Acme software services
Interview
1/25/2026
1/25/2026
View
AI Engineer – Agentic Systems
Acme software services
Shortlisted
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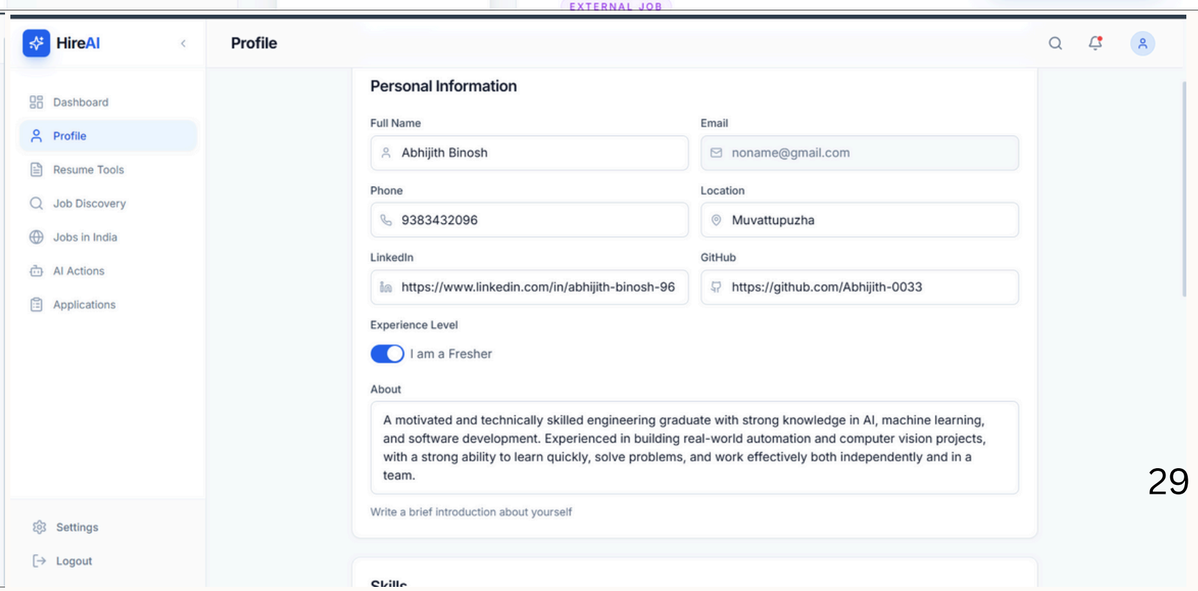
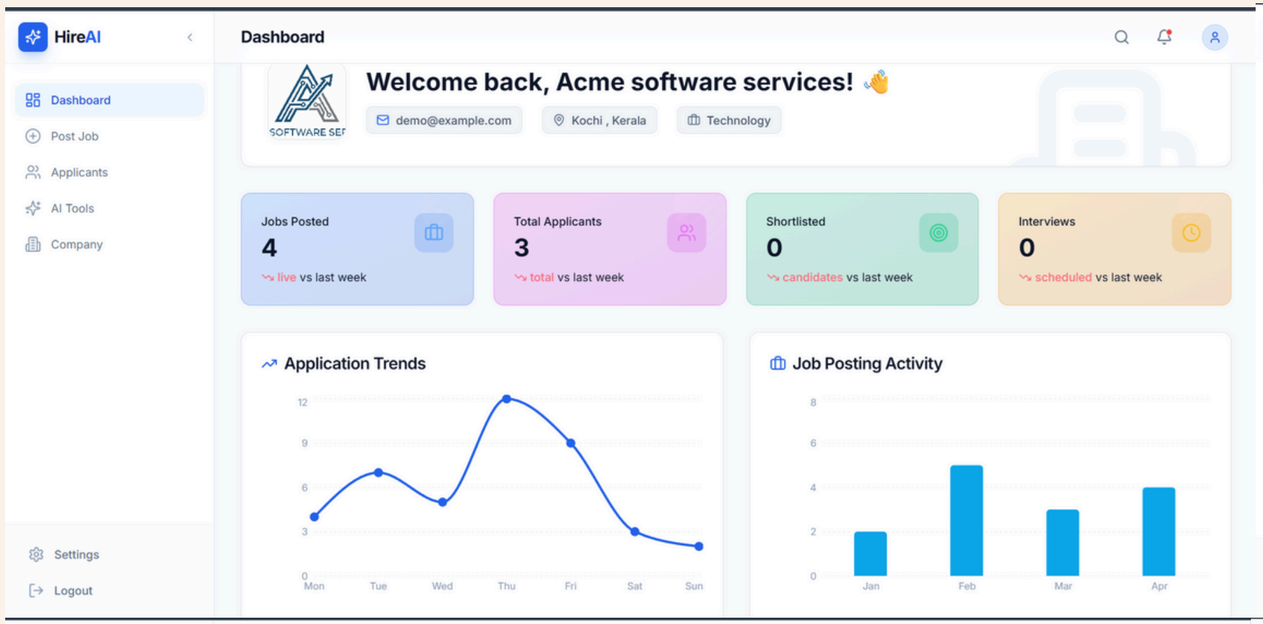
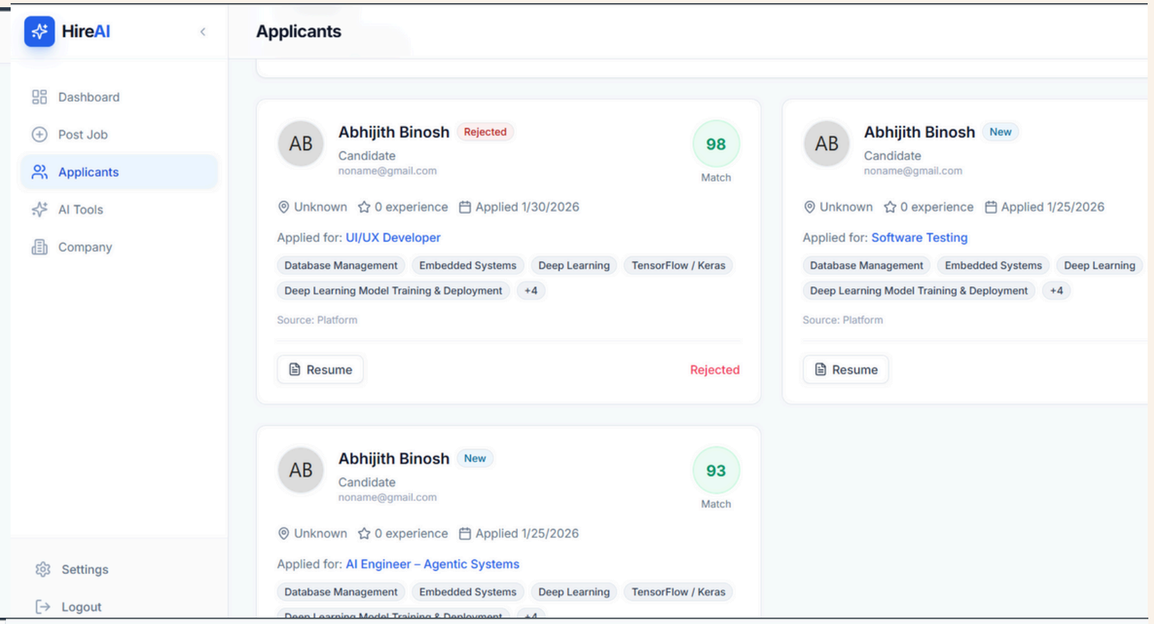
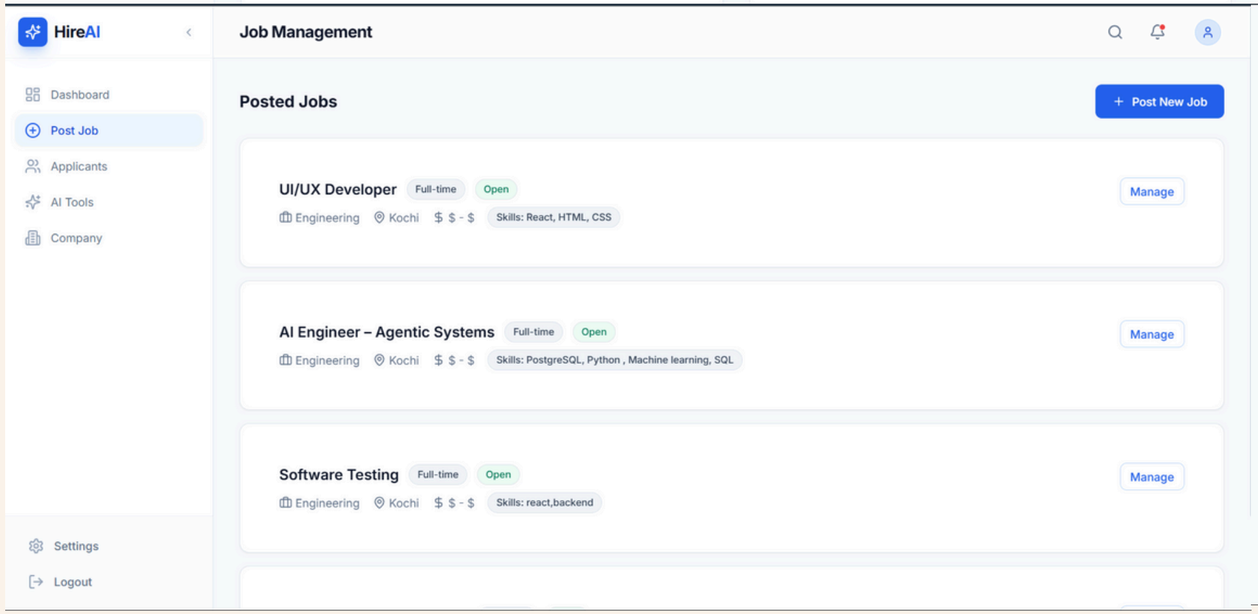
1/25/2026
View

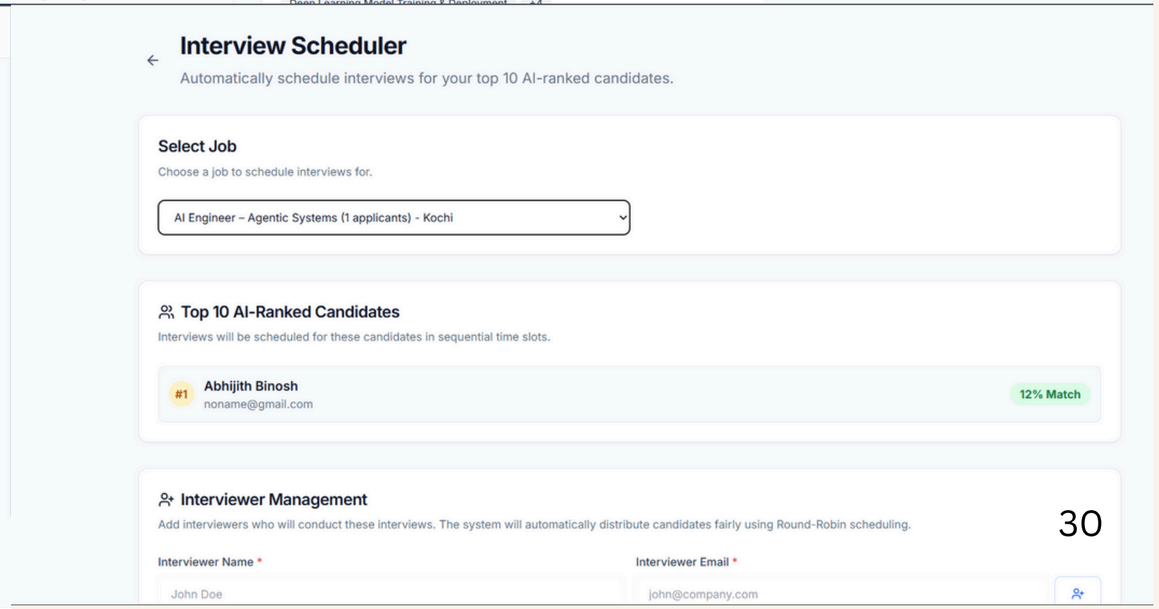

HireAI
Profile
Dashboard
Profile
Resume Tools
Job Discovery
Jobs in India
AI Actions
Applications
Settings
Logout
Personal Information
Full Name
Abhijith Binosh
Email
noname@gmail.com
Phone
9383432096
Location
Muvattupuzha
LinkedIn
https://www.linkedin.com/in/abhijith-binosh-96
GitHub
https://github.com/Abhijith-0033
Experience Level
I am a Fresher
About
A motivated and technically skilled engineering graduate with strong knowledge in AI, machine learning, and software development. Experienced in building real-world automation and computer vision projects, with a strong ability to learn quickly, solve problems, and work effectively both independently and in a team.
Write a brief introduction about yourself

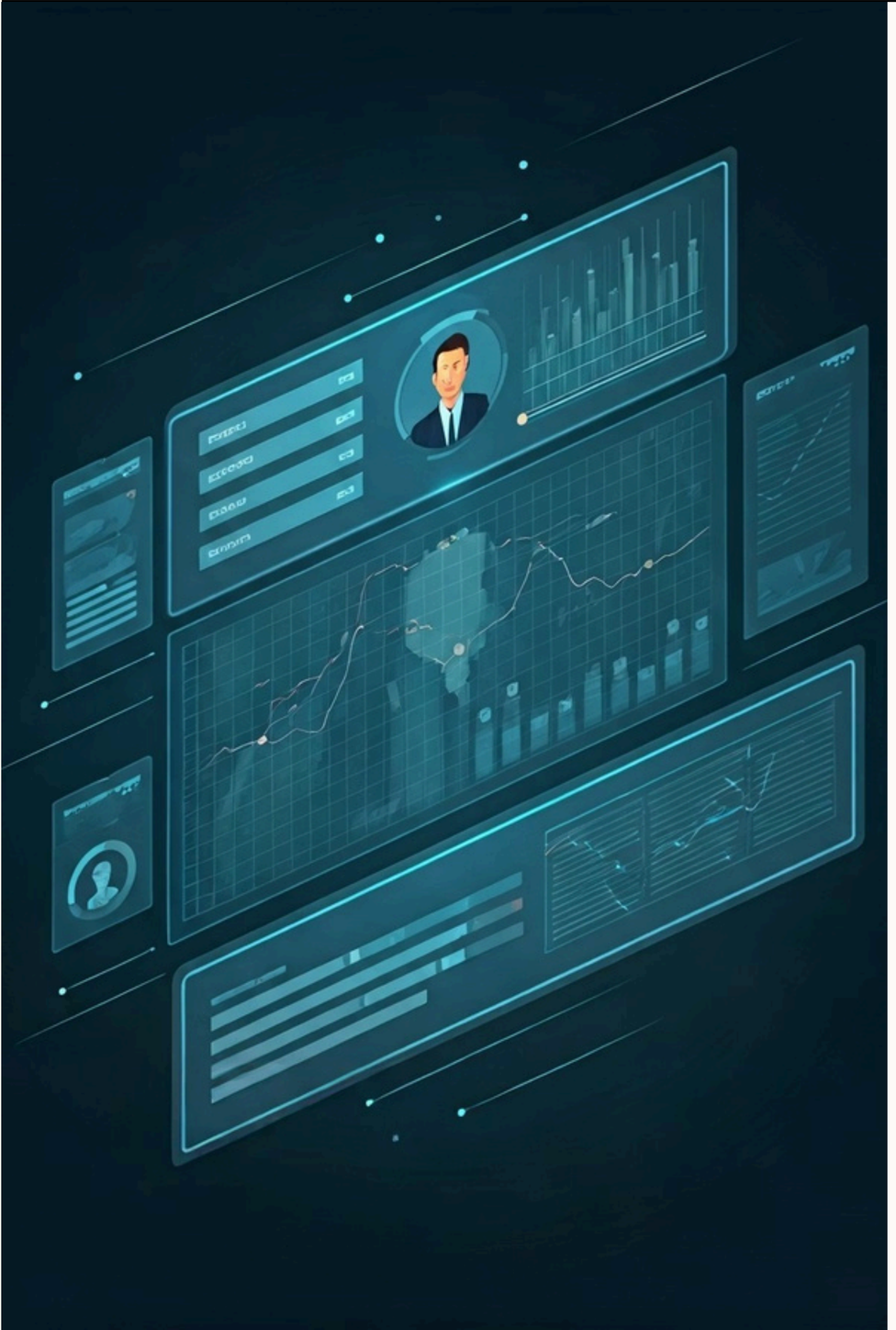
Result


HireAI
Dashboard
Post Job
Applicants
AI Tools
Company
Settings
Logout
Dashboard
Welcome back, Acme software services!
demo@example.com
Kochi , Kerala
Technology
Jobs Posted
4
live vs last week
Total Applicants
3
total vs last week
Shortlisted
0
candidates vs last week
Interviews
0
scheduled vs last week
Application Trends
Job Posting Activity


HireAI
Dashboard
Post Job
Applicants
AI Tools
Company
Settings
Logout
Applicants
Abhijith Binosh
Rejected
Candidate
noname@gmail.com
98
Match
Unknown
0 experience
Applied 1/30/2026
Applied for: UI/UX Developer
Database Management
Embedded Systems
Deep Learning
TensorFlow / Keras
Deep Learning Model Training & Deployment
+4
Source: Platform
Resume
Rejected
Abhijith Binosh
New
Candidate
noname@gmail.com
Unknown
0 experience
Applied 1/25/2026
Applied for: Software Testing
Database Management
Embedded Systems
Deep Learning
Deep Learning Model Training & Deployment
+4
Source: Platform
Resume
Abhijith Binosh
New
Candidate
noname@gmail.com
93
Match
Unknown
0 experience
Applied 1/25/2026
Applied for: AI Engineer – Agentic Systems
Database Management
Embedded Systems
Deep Learning
TensorFlow / Keras


HireAI
Dashboard
Post Job
Applicants
AI Tools
Company
Settings
Logout
Job Management
Posted Jobs
Post New Job
UI/UX Developer
Full-time
Open
Engineering
Kochi
$ - $
Skills: React, HTML, CSS
Manage
AI Engineer – Agentic Systems
Full-time
Open
Engineering
Kochi
$ - $
Skills: PostgreSQL, Python , Machine learning, SQL
Manage
Software Testing
Full-time
Open
Engineering
Kochi
$ - $
Skills: react,backend
Manage


Interview Scheduler
Automatically schedule interviews for your top 10 AI-ranked candidates.
Select Job
Choose a job to schedule interviews for.
AI Engineer – Agentic Systems (1 applicants) - Kochi
Top 10 AI-Ranked Candidates
Interviews will be scheduled for these candidates in sequential time slots.
#1
Abhijith Binosh
noname@gmail.com
12% Match
Interviewer Management
Add interviewers who will conduct these interviews. The system will automatically distribute candidates fairly using Round-Robin scheduling.
Interviewer Name *
John Doe
Interviewer Email *
john@company.com

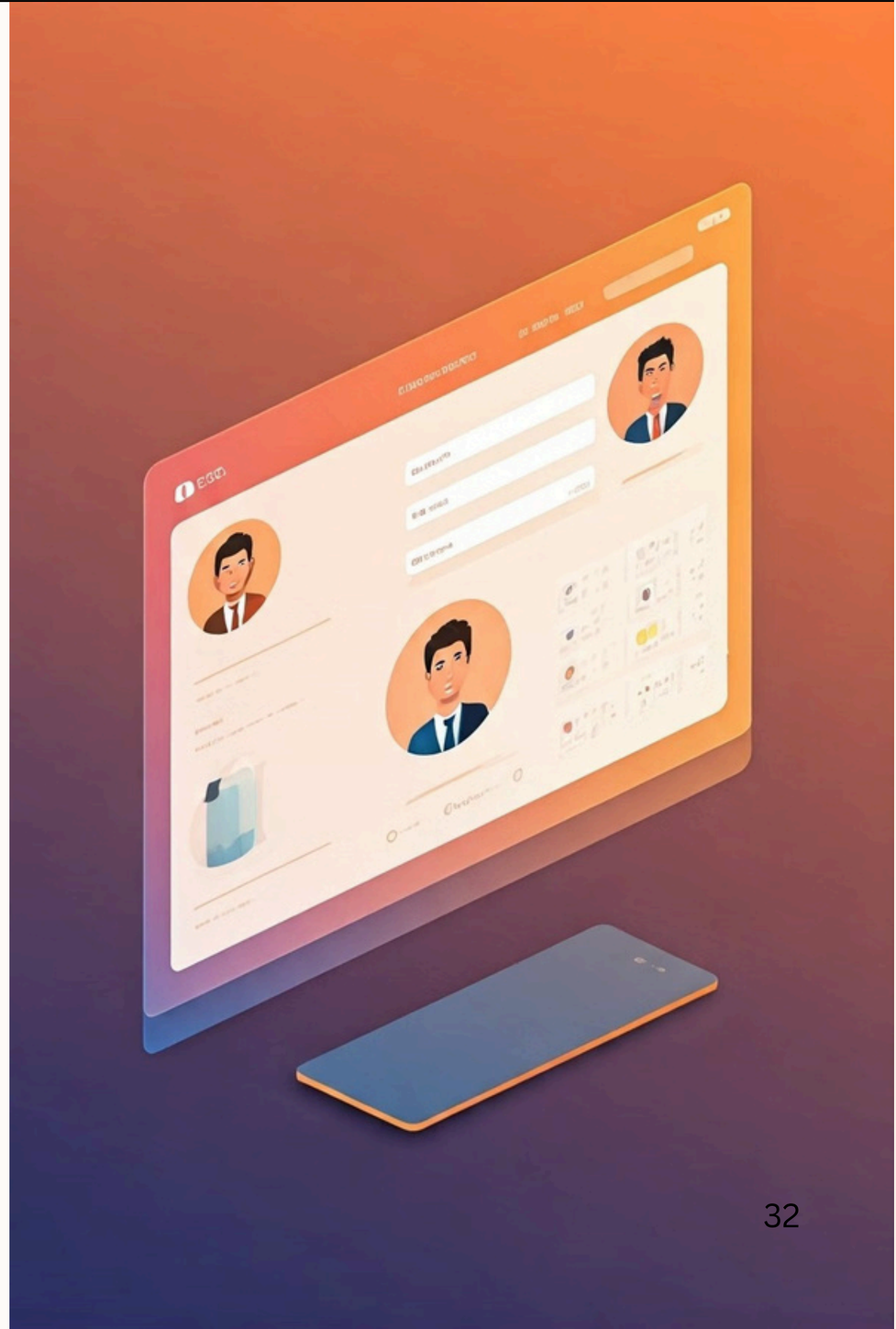


Future Work

- Enhance AI models using advanced machine learning and deep learning techniques for improved matching accuracy.
- Implement full ATS automation to support end-to-end recruitment workflows.
- Integrate advanced talent analytics for predictive hiring insights and performance tracking.
- Improve bias detection and mitigation mechanisms to ensure fair recruitment decisions.
- Expand notification and communication features with full email and calendar integrations.
- Support large-scale deployment with improved scalability and performance optimization.

Conclusion

- HireX presents an AI-powered recruitment platform that automates and streamlines the hiring process.
- The system integrates resume analysis, intelligent matching, candidate ranking, and interview scheduling.
- A modular and scalable architecture enables efficient data management and system extensibility.
- AI-driven decision support improves hiring accuracy, fairness, and efficiency.
- The platform demonstrates strong potential for real-world deployment in modern recruitment environments.



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